cells, so studies of telomeres and telomerase have the promise of providing new insights into conditions such as ageing and cancer. Telomeres prevent the fusion of chromosomal ends with one another.

Chromonema fibre: A chromatin fibre, 100 – 130 nm in diameter, an element of higher order packing within the chromosome. During prophase the chromosomal material becomes visible as very thin filament called chromonemata, which is called as chromatids in early stages of condensation. Chromatid and chromonema are the two names for the same structure a single linear DNA molecule with its associated proteins

Chromomeres: Chromomeres are bead like accumulations of chromatin material which are visible along interphase chromosomes. They can be seen in polytene chromosomes. At metaphase they are not visible.

Types of Chromosomes

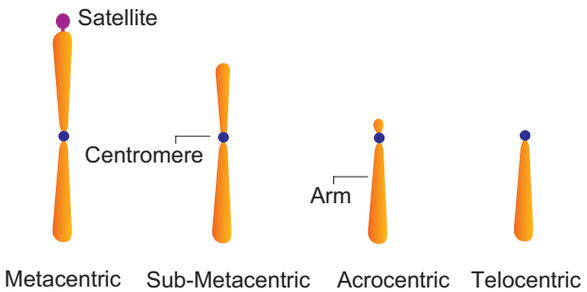


Figure 6.24: Types of chromosomes based on centromere

Based on the position of centromere, chromosomes are called **telocentric** (terminal centromere), **acrocentric** (terminal centromere capped by telomere), **sub metacentric** (centromere subterminal) and **metacentric** (centromere median). The eukaryotic chromosome may be rod shaped (telocentric and acrocentric), L-shaped (sub-metacentric) and V-shaped (metacentric) (Figure 6.24).

Based on the functions of chromosome it can be divided into **autosomes** and **sex chromosomes**.

Autosomes are present in all cells controlling somatic characteristics of an organism. In human diploid cell, 44 chromosomes are autosomes whereas two are sex chromosomes. Sex chromosomes are involved in the determination of sex.

Special types of chromosomes

These chromosomes are larger in size and are called **giant chromosomes** in certain plants and they are found in the suspensors of the embryo. The polytene chromosome and lamp brush chromosome occur in animals and are also called as **giant chromosomes.**

Polytene chromosomes observed in the salivary glands of Drosophila (fruit fly) by **E.G. Balbiani** in 1881. In larvae of many flies, midges (*Dipthera*) and some insects the interphase chromosomes duplicates and reduplicates without nuclear division. A single chromosome which is present in multiple copies form a structure called **polytene chromosome** which can be seen in light microscope. They are genetically active. There is a distinct alternating dark bands and light inter-bands. About 95% of DNA are present in bands and 5% in inter-bands. The polytene chromosome has extremely large puff called **Balbiani rings** which is seen in Chironomous larvae. It is also known as **chromosomal puff**. Puffing of bands are the sites of intense RNA synthesis. As this chromosome occurs in the salivary gland it is known as **salivary gland chromosomes**. Gene expression, transcription of genes and RNA synthesis occurs in the bands along the polytene chromosomes.

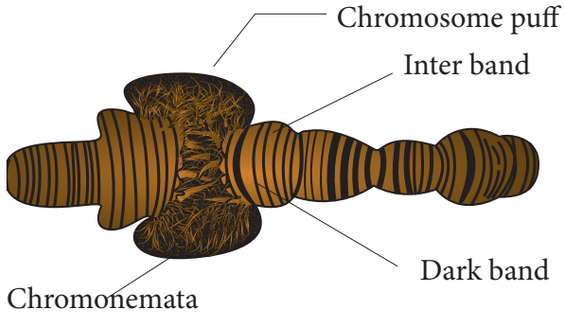


Figure 6.25: Polytene chromosomes

XIth_Botany_Chapter_6_EM.indd 161 5/17/2020 9:10:00 PM